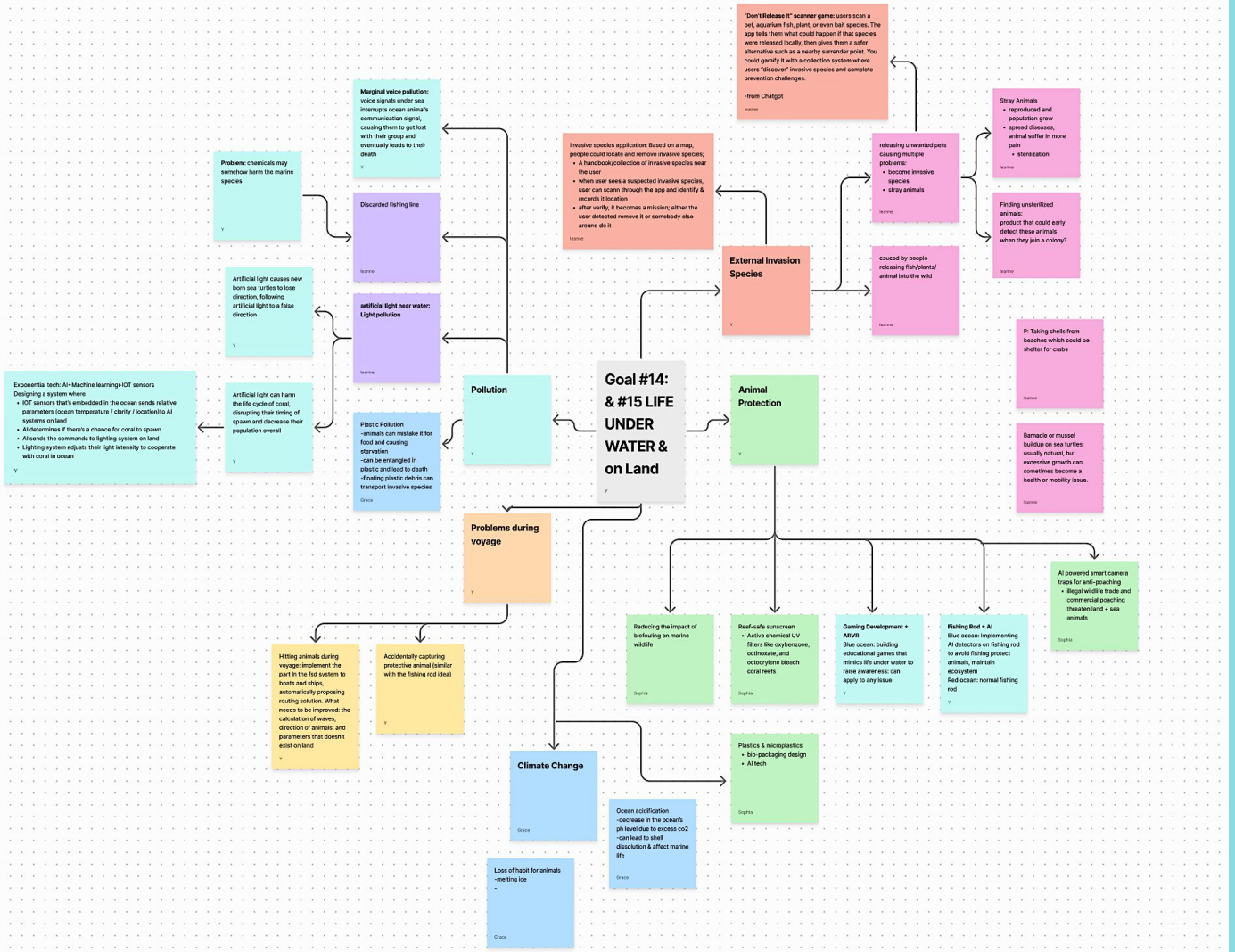
Blue Catch

Grace Li, Sophia Li, Leanne Wang, Daisy Wang





Mind map:

Finding the Problem

Slime Coat Loss
- Handling an accidentally hooked fish with dry hands removes its protective outer slime layer, making it highly vulnerable to diseases and parasites

Grace

Lactic Acid Buildup
- Long, exhausting fights on the line flood the fish's muscles with extreme levels of lactic acid
- can cause delayed mortality.

Grace

Foul Hooking (hooked on body, fins, or eyes outside of the mouth)
- suffers large open wounds that opens them up to bacterial infection

Grace

Deep Hooking (when hooked at esophagus, stomach, or gills)
- can puncture internal organs and cause fatal internal bleeding
- highest mortality rate

Grace

Barotrauma
- Reeling a fish up too fast from deep water expands its swim bladder
- can be fatal or cause severe shock if the fish is not properly decompressed.

Grace

Psychological stress & Environmental factors

Grace

Tissue and Jaw tears
- these injuries decrease the fish's ability to defend and protect itself

Grace

Physical damage

Grace

Competitive products

Y

Problem: fish are only identified after being captured, manual cooperation is needed after the capture is done

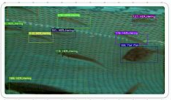
Y

Too expensive, used for commercial scale, not suitable for individual anglers

Y

OTC-SmartFishing
- Camera embedded in fishing net, giving escape tunnels for fish that aren't suppose to be caught

Y



Takeaways: Real-time deliveries to the fisherman + using AI, databases, and models to judge the fish species

Y

The technology behind our solution

Sophia

Problem: Accidental Hooking
- People who fish in the ocean can accidentally catch protected or non-target marine animals
- Even animals are released back to the ocean, the hook can still cause injuries, internal damage, or even death

Ioanne

1. AI Camera on fishhook (AI chip)

2. Small built-in screen

Sophia

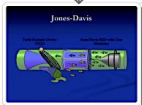


Reasons behind the Problem

Y

competitive products:
- Bycatch reduction for commercial fishing
- Underwater Camera
-

Ioanne





AI camera on the hook:
- a tiny waterproof camera is positioned on the fishing hook
- It captures real time images of animal approaching or interacting ith the bait

Sophia

AI chip:
- an AI chip built into the waterproof camera
- using species recognition technology, it analyzes the animal's physical features and compares them with a database of local species in the area

Sophia

Automatic hook release:
- if a protected species is detected, the fishing rod automatically retracts the hook before the animal can be injured by the sharp hook

Sophia

Built-in screen:
- the fishing rod has a small built-in screen that displays the detected species and provide information about it

Sophia

Even when a hooked protected species is released back into the water alive, it often dies hours or days later (known as discard mortality) due to internal bleeding and infection from embedded metal

Sophia

3D visualization
- Users can see the animal in 3D on the screen along with the species name
- Users can rotate and explore the 3D model
- Adds a new layer of engagement & education for fishers.

Sophia

Invisibility: The angler cannot decide which fish breed to fish and which not to, this creates risks of fishing unwanted ones

Y

Unawareness: The angler aren't aware that the fishing area has protected animals / they aren't aware that the fishing bait can attract protected animals

Y

Potential solution: upload fishing data from the Ai chip to a community platform, raising awareness for the public + help them to identify the location for which the protected fish may occur

Y

Mind map:
Defining the Problem

Define Challenge & Market Opportunity

- **Accidental bycatch:** Recreational anglers can unintentionally hook protected animals because they cannot always see what is approaching their bait underwater.
  - ❖ Although people could release them immediately back to the ocean, hooks can injure the mouth, head, or digestive tract. Swallowed hooks can interfere with feeding and cause starvation and death.
- NOAA states that accidental capture or injury of sea turtles and marine mammals during fishing is a significant threat to their health and recovery.
- **Large potential market:** In the U.S. alone, 12.7 million people participated in saltwater fishing in 2022, making approximately 104 million saltwater fishing trips.



Exponential Drivers

Artificial intelligence (AI)

- Rapid advances in computer vision are making real-time species recognition faster and more accurate based on physical features (ex. body shape, color, size)
- Smaller and more powerful AI chips allow complex image recognition to run directly on compact devices with minimal delay

Miniaturized camera technology

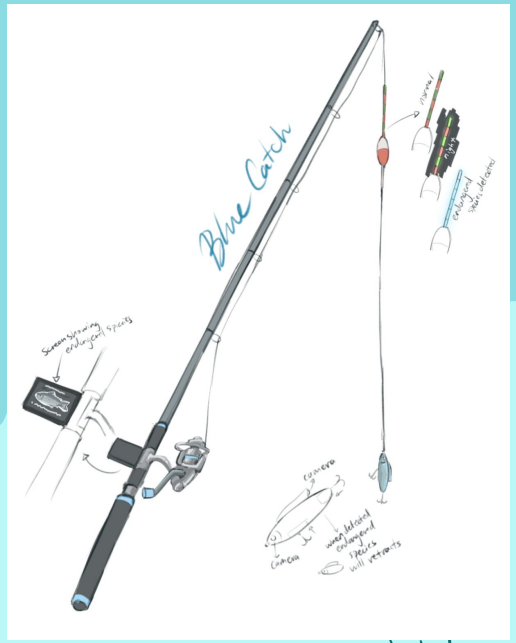
- Cameras are becoming smaller and lighter while maintaining high image quality
- Tiny waterproof cameras can now be integrated into fishing equipment without affecting usability

Real-time 3D visualization

- Improvements in 3D rendering technology make interactive visualizations more realistic
- Advances in computing power allows detailed 3D models to run on everyday devices
- Interactive visualization adds a new layer of engagement & educational content

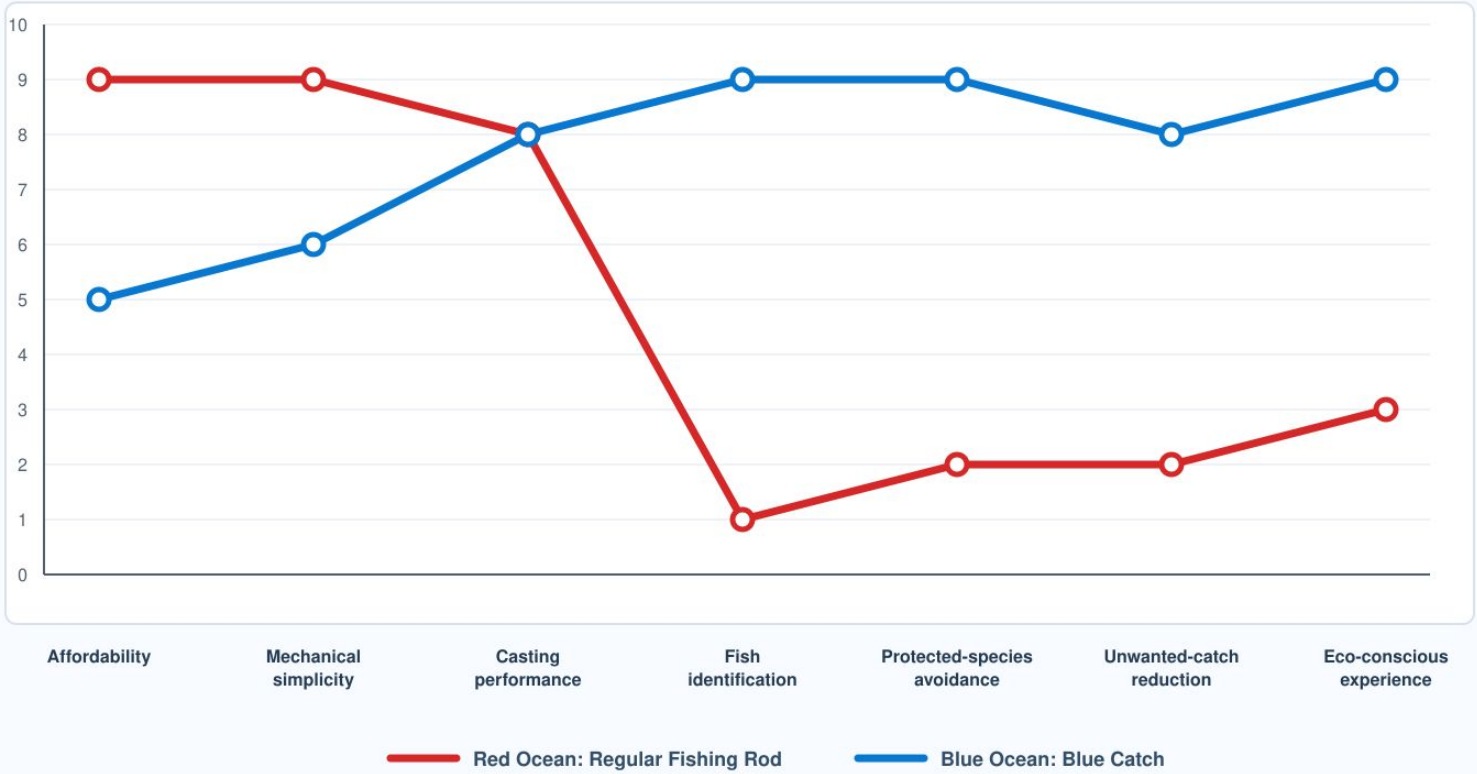
Our Solution

- **Smart fishing equipment**
  - **Camera on fish-shaped bait with retractable hooks**
    - Includes AI and when spotting a fish nearby, the AI will compare it to endangered species to determine whether that fish is one
    - If endangered then the hooks will retract into the bait to avoid injuring the fish immediately
    - Hooks will remain retracted until that fish had left
  - **Screen on the pole**
    - Shows the endangered species captured as an interactive 3D model, along with the photo camera just took with fun facts about it
  - **Floats equipped with lights that can change color**
    - Will light up when hooks are retracted to notify the user
    - If at night will change color since the float lights up at night already



Blue Ocean Strategy Value Map

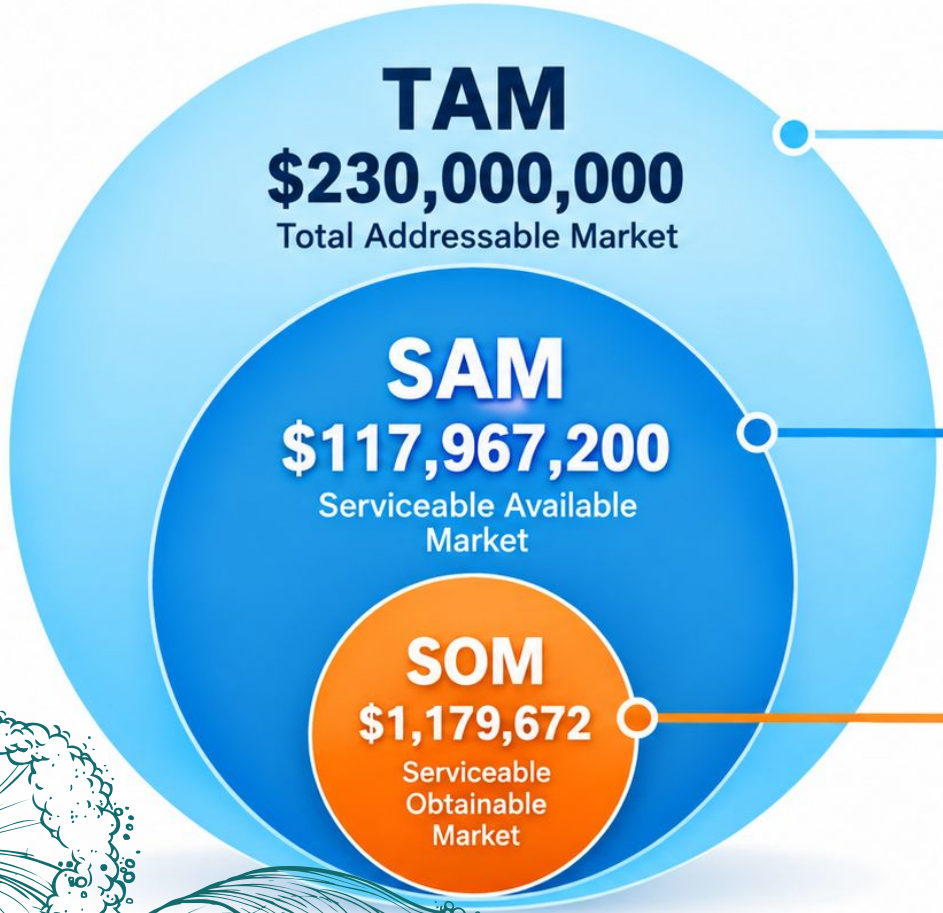
Regular Fishing Rod vs Blue Catch



A7 Market Potential

Leanne, Daisy, Sophia, Grace

TAM / SAM / SOM Market Potential



TAM – Total Addressable Market

$230,000,000

Based on U.S. recreational and commercial fishing assumptions.

SAM – Serviceable Available Market

$117,967,200

Based on 52% serviceability / integration feasibility.

SOM – Serviceable Obtainable Market

$1,179,672

Based on 1% market capture assumption.

Interview Feedbacks

#1 Interviewee: professional fisher / Interview date: September 26, 2026

- Agreed the idea of using a camera and AI to identify fish before they are hooked is really useful
- He liked that the system would immediately retract the hooks when it detects a protected species rather than relying on humans to recognize the animal after the damage already happened.
- However, he was concerned about how accurately the AI could identify fish underwater, especially in low-light conditions. In addition, he was more interested if the product offered everyday fishing benefits.

#2: Interviewee: occasional fisher / Interview date: September 27, 2026

- The interviewee doesn't fish too often, but she liked how the concept adds a layer of education to fishing.
- She thought showing the protected fishes on screen with 3D models and fun facts could make fishing more interesting for people who are not experienced anglers.
- concerned that the technology might be complicated for people who are not familiar with more technical devices (such as seniors), so she suggested making the interface simple and easy to understand.

#3: Interviewee: amateur fishing hobbyist / Interview date: September 29, 2026

- Passionate about fishing and is interested in the concept. He has personal experience on the issue, where he had accidentally caught an endangered species while fishing and had to set it free, though it was already injured and likely had a hard time surviving afterwards.
- Interviewee also mentioned that he would prefer to see the camera's view on the screen, as it would help him see if there is actually any fish in that location.

TAM - Total Addressable Market

As our products currently targets to U.S market, number of people who participates in the recreational fishing industry and the number of people who employs in the commercial fishing industry is taking into accountability for the calculation of the total addressable market.

- 57 Million U.S. Citizens participated in recreational fishing until 2025
- 51,902 U.S. Citizens are employed in the commercial fishing industry until 2025

Assumptions:

- Assuming 20% of the people who once fished is an amateur angler
- Assuming 10% of the amateur angler would purchase our product
- Assuming 30% of the commercial angler would purchase our product
- Assuming all customer would spend 199$ annually on our product (the retail price of a fishing rod)

Calculations:

TAM = (5700,000,0 * 0.2 * 0.1 + 51902 * 0.3) * 199 = **$ 230,000,000**

SAM - Serviceable Available Market

Total number of potential customers: 1.14 million - Established based on previous assumption
Average Annual Revenue per Customer: $199 - The retail price of a single fishing rod
Percentage of Customers we can serve:
The statistics of serviceable customers in the market is highly dependent on the availability of the technology. Our product makes usage of a combination of three types of exponential technology: Artificial Intelligence, Miniaturized camera technology, and Real-time 3D visualization
Assumptions:

- **Existing technology of AI tool** - Fishial - has an identification ability of detecting over 865 types of fish, while real life oceanic environment is still needed to be trained. Percentage availability: 70%
- Waterproof, high resolution, and mini **underwater camera** is highly available, but short delay of image still needs to be explored. Percentage availability: 76%
- Mature technology like Apple AR Quick look can demonstrate **3D models**. Percentage availability: 90%

Calculation: (70% + 76% + 90%) / 3 = 52%
SAM = 1,140,000 customers * $199 * 52% = **$ 117,967,200**

SOM - Service Obtainable Market

Number potential customers: 1.14 million / **Average Annual Revenue per Customer:** $199

Percentage of Customers we can serve: 52%

Percentage of market you can capture:

The capture rate is calculated by using the percentage of potential customers who has the desire to purchase over the total number who had the opportunity. To this end, based on our estimated production, distribution, and early adoption capacity, we assume that **5,000 customers** would purchase our product annually in the first three years. Therefore, percentage of market captured is calculated by:

5000 / 592800 = 0.84%

Rounding up the percentage to 1%, the number for our SOM is:

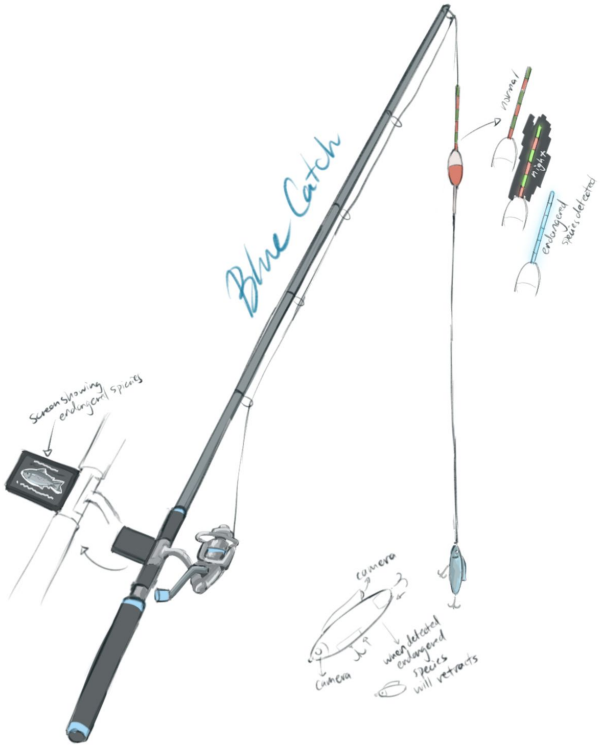
SOM = 1,140,000 * 52% * 1% * $199 = **$1,179,672**

Competitive Analysis

Daisy, Sophia, Leanne, Grace

Product Definition

Blue Catch is an AI-powered smart fishing rod designed to help anglers avoid accidentally hooking protected marine animals. It uses a camera attached to fish-shaped bait to identify nearby fish and automatically retracts its hooks when a protected species is detected. A screen on the fishing rod displays the animal's photo, an interactive 3D model, and fun facts. Our mission is to protect marine life while making fishing safer and more educational.

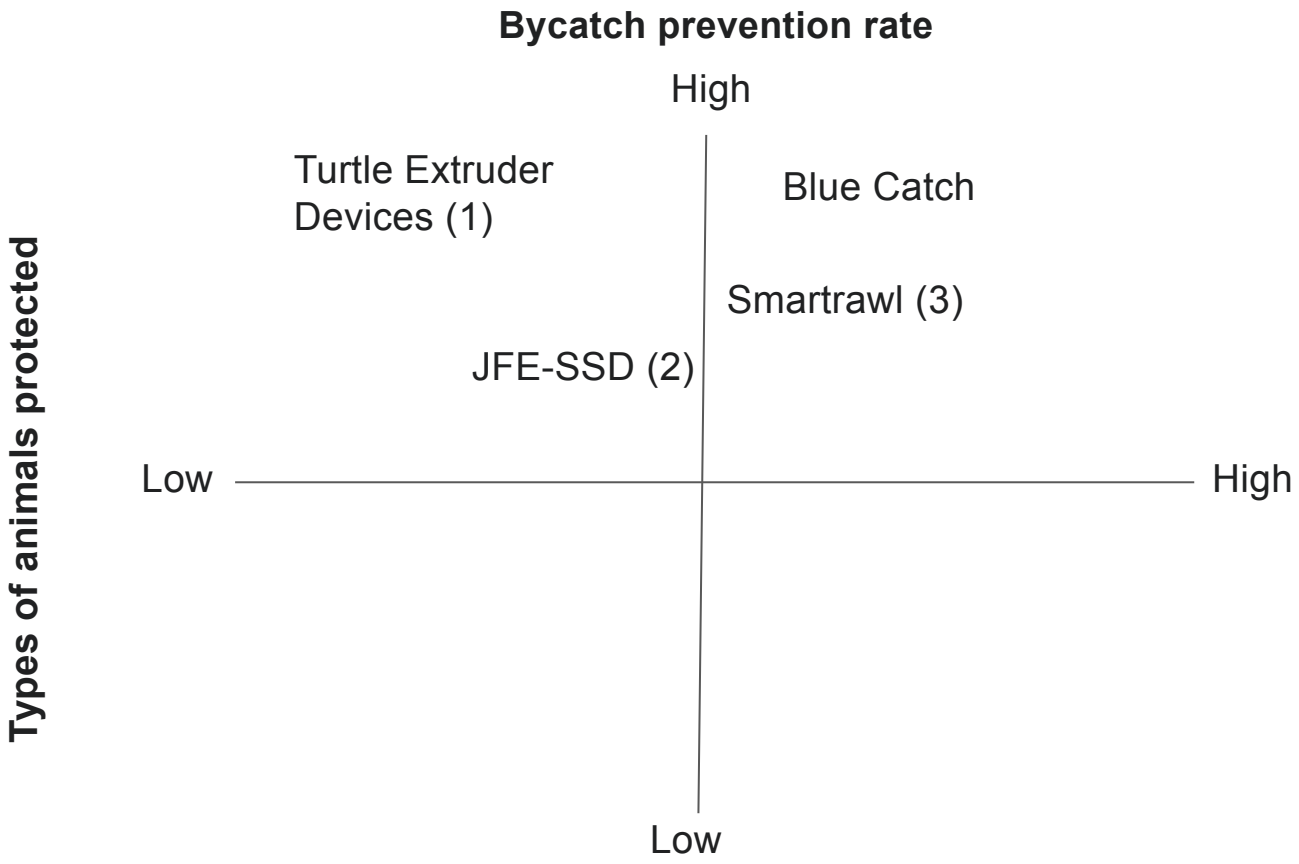


SWOT Analysis of BlueCatch

Strength (S)	Weakness (W)
• **Clear and effective conservation purpose** - with bycatch being a major threat to the recovery of protected species, the BlueCatch product provides an experimental solution to this **(1)** • **0 Latency** - Unlike other substitute products that have lag or identify the species after being caught, BlueCatch aims for 0 latency and warns anglers before catching fish **(2)** • **3D Modeling technique** - BlueCatch not only helps anglers avoid catching, but also provides a 3D model that helps them learn about new species **(3)** • **Cost-effective** - The chip sends signals from underwater instead of uploading video, saving battery life which saves consumer cost **(4)**	• **Underwater visibility can be unreliable** - ocean waves, changing light sources, and changing water conditions can reduce camera detection accuracy. **(5)** • **Larger float** - since Bluecatch embeds AI-powered chips within and waterproofs sensors and processors, the float would be larger compared with regular ones, which could decrease the chance of successful fishing. **(6)** • **Species limit** - unrecorded species would be hard to detect and thus may still cause accidental bycatch **(7)**
Opportunities (O)	**T (Threats)**
• **Large consumer base** - Approximately 57 million Americans participate in fishing in 2025, and more can be expected inthe following years; a large TAM can be expected **(8)** • Rising awareness for ocean conservation - public and governmental demand for conservation tools and strategy shows potential for product sales. **(9)** • Market opportunity beyond the fishing industry: similar techniques can be applied on a larger scale in aquaculture sites or marine monitoring projects worldwide. **(10)**	• Competitors in the industry - existing underwater camera and ocean monitoring firms may possess similar features in their products that are easier to implement on a larger scale **(11)** • False alarms or missing detection may decrease user trust - if users no longer trust the product, sales will go down **(12)** • Privacy concerns - the chip may record location information and species information; leaking such information would also raise trust issues.**(13)**

1) https://www.fisheries.noaa.gov/topic/bycatch
2) https://www.fisheries.noaa.gov/insight/understanding-bycatch
3) Vision and function design of BlueCatch
4) Technical function design of BlueCatch
5) https://www.sciencedirect.com/science/article/pii/S1570870526002982
6) https://pmc.ncbi.nlm.nih.gov/articles/PMC12157327/
7) https://www.sciencedirect.com/science/article/pii/S1570870526002982
8) https://www.statista.com/topics/1163/recreational-fishing/#topicOverview
9) https://www.fisheries.noaa.gov/insight/understanding-bycatch
10) https://www.fisheries.noaa.gov/topic/laws-policies
11) https://www.thuenen.de/en/institutes/baltic-sea-fisheries/projects/fisheries-and-survey-technology/otc-smart-fishing-sensorsystem-fuer-smarte-fischernetze
12) Anticipated social trust issue in BlueCatch
13) https://action.safe.ai/news/the-chip-security-act-separating-fact-from-fiction

Competitive Landscape



Sources:

1. https://www.fisheries.noaa.gov/southeast/bycatch/turtle-excluder-devices

2. https://pmc.ncbi.nlm.nih.gov/articles/PMC3696171/

3. https://thenationalrobotarium.com/ai-empowered-fishing-net-to-help-prevent-marine-bycatch/

https://ecomagazine.com/news/fisheries-aquaculture/smartrawl-ai-empowered-fishing-net-to-help-prevent-marine-bycatch/

Comparison Chart

	Turtle Excluder Device (2)	Smartrawl (1)	JFE-SSD (3)	Blue Catch
Smart identification/ sorting	❌	✅	❌	✅
Preventing bycatch	✅	✅	✅	✅
Underwater vision (through camera)	❌	✅	❌	✅
3D model & educational purposes	❌	❌	❌	✅

sources:

1:

https://fisorg.uk/wp-content/uploads/2025/06/Smartrawl-Phase-6-Report-Final-copy_2.pdf

https://marineguardian.eu/smartrawl-tackles-bycatch-of-species/

2:

https://www.fisheries.noaa.gov/feature-story/gearing-turtle-excluder-devices-strengthen-fisheries-and-help-recover-sea-turtle

https://www.fisheries.noaa.gov/southeast/bycatch/turtle-excluder-devices

3:

https://www.researchgate.net/publication/242019260_Investigations_on_juvenile_fish_excluder_cum_shrimp_sorting_device_JFE-SSD

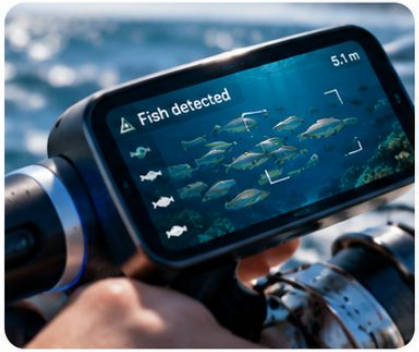
UVP/VVP of BlueCatch

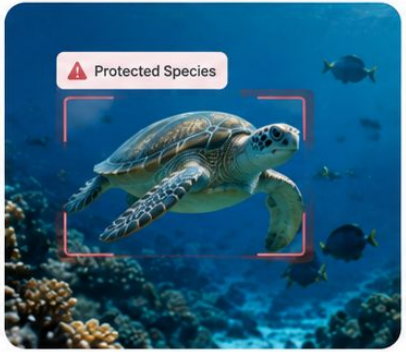
Daisy, Sophia, Leanne, Grace

Value Proposition

For anglers who want to avoid accidentally hooking protected marine animals and learn more about marine life, Blue Catch is an AI-powered smart fishing rod that identifies nearby fish, automatically retracts its hooks when protected species are detected, and displays the animal's photo, interactive 3D model, and fun facts to help prevent injuries to marine life while making fishing safer and more educational.









Fish smarter, protect what you love.

Blue Catch is an AI-powered smart fishing rod that identifies nearby fish, automatically retracts its hooks when protected species are detected, and displays the animal's photo, interactive 3D model, and fun facts — helping you fish safer while learning more about marine life.

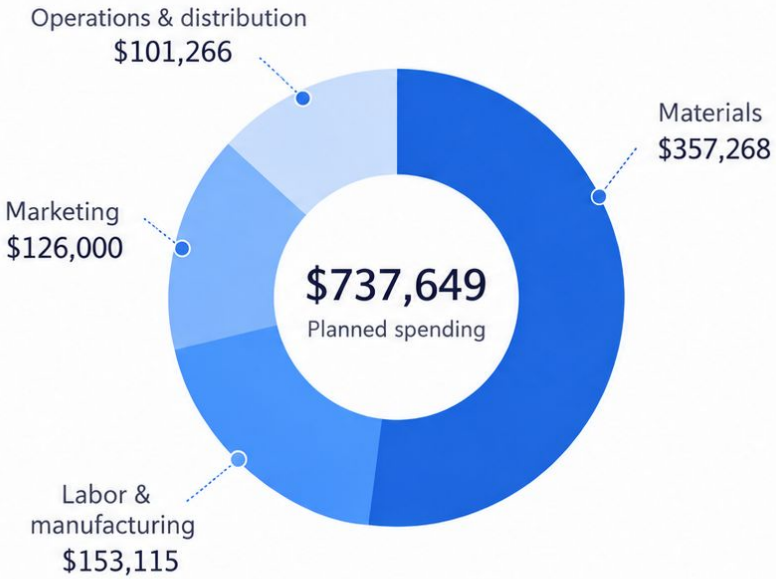
EXPLORE BLUE CATCH ›

Guerilla Financial Pro Forma

https://docs.google.com/spreadsheets/d/1PcS-zDE94jpFI5VOcXDihgA-AtQnOXyXWg-bBoQf8Xo/edit?usp=sharing

Use of Funds

Blue Catch | 2025 projected spending



Based on estimated annual cash expenses. Excludes tax, depreciation and amortization.

Materials

Cameras, AI chips, screens and rod components

Labor & manufacturing

Rod assembly and system integration

Marketing

Online advertising and offline promotions

Operations & distribution

Rent, storage, packaging, shipping and fees

Labor & manufacturing: Labor $102,076.52 + Factory $51,038.26 = $153,114.78
Operations & distribution: Storage $18,000 + Rent $36,000 Packaging $1,880.36 + Shipping $32,906.25 Interest $12,000 + Platform $479.88 = $101,266.49